Regime-Adaptive Portfolio Optimization via Causal Hidden Markov Models

Drawdown Control Through Regime-Conditional Convex Optimization

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Abstract

We present a regime-adaptive portfolio framework that combines a Gaussian Hidden Markov Model (HMM) for market regime detection with regime-conditional convex optimization. The key methodological contribution is the replacement of standard Viterbi decoding and forward-backward smoothing with a strictly causal forward filter, eliminating within-window lookahead bias present in most HMM-based portfolio systems. Applied to eight liquid ETFs over 2006–2024, the strategy achieves a maximum drawdown of −26.1% versus −60.4% for SPY buy-and-hold, with a full-period Sharpe ratio of 0.453 versus 0.433. A frozen out-of-sample evaluation — training exclusively on pre-2019 data — yields a Sharpe of 0.324, confirming that performance does not depend on lookahead in the walk-forward procedure. Bootstrap significance testing yields $p = 0.376$, indicating that excess Sharpe over SPY is not statistically significant; the strategy's primary value proposition is tail risk control rather than return generation.

1 Introduction

Financial markets exhibit distinct statistical regimes — periods of low volatility trending growth, high volatility mean reversion, and acute crisis episodes — that are not well captured by time-invariant models. Portfolio strategies optimized under a single distributional assumption perform poorly when the underlying regime shifts, as evidenced by the catastrophic drawdowns of traditionally diversified portfolios during the 2008 Global Financial Crisis (GFC) and the 2022 simultaneous equity and bond selloff.

Hidden Markov Models [Hamilton, 1989] provide a principled framework for modeling latent market states. In the portfolio management literature, HMM-based regime detection has been applied to asset allocation [Ang and Bekaert, 2002], volatility forecasting [Turner et al., 1989], and risk management [Guidolin and Timmermann, 2007]. However, most implementations rely on the Viterbi algorithm or forward-backward smoothing for state inference — both of which use future observations within the estimation window, introducing a subtle but consequential form of lookahead bias.

This paper makes the following contributions:

1. **Causal forward filter**: We replace Viterbi decoding with a strictly causal forward filter that computes $P(\text{state}_t \mid \mathbf{o}_{1:t})$ using only past observations, eliminating within-window lookahead bias.

2. **Posterior probability blending**: Rather than hard regime assignment, we blend optimizer outputs across all regimes weighted by filtered posterior probabilities, reducing turnover at transition boundaries.

3. **Nested cross-validation**: State count selection via BIC is performed independently at each quarterly fold, preventing information leakage from the full sample into model selection.

4. **Comprehensive validation**: We provide a frozen out-of-sample evaluation, feature ablation study, walk-forward leakage audit, and bootstrap significance test — a level of methodological scrutiny absent from most published backtests.

The remainder of this paper is organized as follows. Section 2 describes the full methodology. Section 3 details the data and implementation. Section 4 presents performance results. Section 5 reports robustness analyses. Section 6 discusses limitations, and Section 7 concludes.

2 Methodology

2.1 Feature Construction

We construct three features from daily asset returns:

$$r_t^{\text{SPY}} = \log\left(\frac{P_t^{\text{SPY}}}{P_{t-1}^{\text{SPY}}}\right) \tag{1}$$

$$\sigma_t = \sqrt{\frac{252}{W_\sigma}\sum_{i=0}^{W_\sigma-1}(r_{t-i}^{\text{SPY}})^2} \tag{2}$$

$$\bar{\rho}_t = \frac{1}{N(N-1)}\sum_{i\neq j}\hat{\rho}_{ij,t} \tag{3}$$

where $W_\sigma = 21$ trading days, $\hat{\rho}_{ij,t}$ is the rolling $W_\rho = 63$-day pairwise correlation between assets i and j, and $N = 8$ is the number of assets. Both window parameters are configurable. Features are standardized to zero mean and unit variance before HMM fitting.

2.2 Gaussian Hidden Markov Model

We model the feature sequence $\{\mathbf{x}_t\}$ as observations from a K-state Gaussian HMM:

$$P(\mathbf{x}_t \mid s_t = k) = \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{4}$$

$$P(s_t = j \mid s_{t-1} = i) = A_{ij} \tag{5}$$

where $s_t \in \{1, \ldots, K\}$ is the latent regime, $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$ are regime-specific mean and covariance, and A is the $K \times K$ transition matrix. Parameters $\{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, A, \boldsymbol{\pi}\}$ are estimated by the Baum-Welch algorithm (EM). Multiple random initializations ($n_{\text{init}} = 10$) are used to mitigate local optima; the initialization with the highest log-likelihood is retained.

2.3 State Count Selection via BIC

The number of states K^* is selected at each quarterly fold by minimizing the Bayesian Information Criterion:

$$\text{BIC}(K) = -2\ell(\hat{\theta}_K) + p_K \log T \tag{6}$$

where $\ell(\hat{\theta}_K)$ is the maximized log-likelihood, T is the number of training observations, and p_K counts the free parameters:

$$p_K = K(K-1) + Kd + K\frac{d(d+1)}{2} + (K-1) \tag{7}$$

with $d = 3$ features. Candidates $K \in \{2, 3, 4\}$ are evaluated independently at each fold.

2.4 Causal Forward Filter

Standard HMM inference tools compute smoothed posteriors $P(s_t \mid \mathbf{x}_{1:T})$ using forward-backward passes over the full observation sequence. This introduces within-window lookahead: the regime label assigned to day t depends on observations $\mathbf{x}_{t+1}, \ldots, \mathbf{x}_T$, which are unavailable in real-time.

We replace smoothing with a forward-only filter. The filtered posterior $\alpha_t(k) = P(s_t = k \mid \mathbf{x}_{1:t})$ is computed recursively:

$$\alpha_1(k) = \pi_k \cdot p(\mathbf{x}_1 \mid s_1 = k) \tag{8}$$

$$\alpha_t(k) = p(\mathbf{x}_t \mid s_t = k) \sum_{j=1}^{K} \alpha_{t-1}(j) A_{jk} \tag{9}$$

with normalization $\alpha_t \leftarrow \alpha_t / \sum_k \alpha_t(k)$ at each step to prevent underflow. The regime assignment on day t depends exclusively on $\mathbf{x}_{1:t}$.

Performance impact: Replacing smoothed posteriors with filtered posteriors reduces full-sample Sharpe from 0.562 to 0.453. This degradation is expected and desirable — Viterbi's within-window smoothing assigns better labels in hindsight. The forward-filter numbers reflect what a truly causal system would have achieved.

2.5 Walk-Forward Retraining

To prevent the HMM from using future market structure in regime detection, we employ quarterly walk-forward retraining with an expanding training window:

1. At each quarterly retraining date τ_q, fit the HMM on all data $\mathbf{x}_{1:\tau_q}$

2. Apply the fitted model via forward filter to observations in the subsequent quarter $[\tau_q + 1, \tau_{q+1}]$

3. Advance to the next quarter and repeat

Train and test windows are non-overlapping by construction. A programmatic audit confirms zero leakage across all 74 quarterly folds (Section 5).

2.6 Label Permutation Handling

HMM states have no intrinsic semantic identity — state index 0 in quarter q need not correspond to the same market condition as state index 0 in quarter $q+1$. We resolve this by assigning regime labels based on mean return rank after every retrain: the state with the lowest mean SPY return is labeled Crash, followed by Bear, Sideways, and Bull. This mapping is deterministic and requires no cross-quarter state matching.

2.7 Regime-Conditional Optimization

Four convex optimization programs are defined, one per regime:

Bull — Maximum Sharpe: Using the Lagrangian auxiliary variable reformulation [Cornuéjols and Tütüncü, 2006]:

$$\min_{\mathbf{y}} \quad \mathbf{y}^\top \hat{\Sigma} \mathbf{y} \tag{10}$$

$$\text{s.t.} \quad (\hat{\boldsymbol{\mu}} - r_f)^\top \mathbf{y} = 1, \quad \mathbf{y} \geq \mathbf{0} \tag{11}$$

where $\mathbf{w} = \mathbf{y}/\mathbf{1}^\top \mathbf{y}$, r_f is the time-varying 3-month T-bill rate (`^IRX`), and $\hat{\Sigma}$ is the Ledoit-Wolf shrinkage estimator [Ledoit and Wolf, 2004].

Bear — Risk Parity: Weights equalize each asset's contribution to portfolio variance:

$$\min_{\mathbf{w}} \quad \mathbf{w}^\top \hat{\Sigma} \mathbf{w} - \frac{1}{N} \sum_{i=1}^{N} \log w_i, \quad w_i \geq 0.01 \tag{12}$$

Sideways — Minimum Variance:

$$\min_{\mathbf{w}} \quad \mathbf{w}^\top \hat{\Sigma} \mathbf{w} \tag{13}$$

$$\text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq 0.60 \tag{14}$$

Crash — Inverse-Volatility Safe Haven: Equal risk contribution over the safe-haven subset $\mathcal{S} = \{\text{IEF, TLT, GLD}\}$:

$$w_i = \frac{\sigma_i^{-1}}{\sum_{j \in \mathcal{S}} \sigma_j^{-1}}, \quad i \in \mathcal{S} \tag{15}$$

where σ_i is the trailing volatility of asset i.

2.8 Posterior Probability Blending

Rather than committing to a single regime optimizer, daily weights are computed as a convex combination weighted by filtered posteriors:

$$\mathbf{w}_t = \sum_{k \in \mathcal{K}} \alpha_t(k) \cdot \mathbf{w}_t^{(k)} \tag{16}$$

where $\mathbf{w}_t^{(k)}$ is the weight vector produced by regime k's optimizer using all returns available through day t. Blending reduces turnover at transition boundaries and propagates the HMM's regime uncertainty into the portfolio allocation.

All four optimizer weight vectors are recomputed together whenever the dominant regime changes or at each quarterly retraining date, ensuring consistency across the blending.

3 Data & Implementation

3.1 Asset Universe

The investable universe consists of eight liquid US-listed ETFs spanning major asset classes:

Table 1: Asset Universe

Ticker	Description	Class
SPY	S&P 500	Equity
QQQ	NASDAQ-100	Equity
EFA	MSCI EAFE	Intl Equity
VNQ	US REITs	Real Estate
TLT	20+ Year Treasury	Long Bond
IEF	7-10 Year Treasury	Mid Bond
LQD	Investment Grade Corp	Credit
GLD	Gold	Commodity

Daily adjusted closing prices are sourced from Yahoo Finance via `yfinance`. The sample covers January 2005 through December 2024, yielding 5,031 trading days after alignment. Log returns are computed as $r_t = \log(P_t/P_{t-1})$.

3.2 Evaluation Design

The sample is split into a training period (January 2006 – December 2018) and a held-out test period (January 2019 – December 2024). The training period encompasses the Global Financial Crisis, the European debt crisis, and the post-GFC bull market. The test period covers the COVID-19 crash, the 2020–2021 equity recovery, and the 2022 simultaneous equity and bond selloff.

Walk-forward retraining uses a quarterly frequency (`QS` in pandas date offset notation) with an expanding training window — the minimum training window is 252 days. The walk-forward procedure is applied across the full 2006–2024 period; the 2019–2024 slice is used as the held-out evaluation period.

3.3 Transaction Costs

A flat round-trip cost of 2 basis points is applied to all portfolio weight changes:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot \sum_i |w_{t,i} - w_{t-1,i}| \tag{17}$$

where $c = 0.0002$. This is conservative for liquid ETFs at institutional scale. Average daily turnover is 4.28%, implying an annualized transaction cost of approximately 0.22%.

3.4 Benchmarks

We compare against five benchmarks, all net of the same 2bp transaction cost model:

- **SPY**: Buy-and-hold S&P 500
- **Equal Weight**: Monthly-rebalanced equal allocation across all 8 assets
- **60/40**: 60% SPY / 40% IEF, monthly rebalanced
- **Momentum**: Monthly rebalancing to the top-4 assets by trailing 12-month return
- **Risk Parity**: Static risk parity over all 8 assets

3.5 Risk-Free Rate

The time-varying risk-free rate is sourced from the 3-month US Treasury Bill yield (`^IRX` via Yahoo Finance). It is used both in the maximum-Sharpe QP objective and in all reported Sharpe ratio calculations, ensuring consistency between optimization inputs and performance reporting.

3.6 Implementation

The full pipeline is implemented in Python using `hmmlearn` [hmmlearn developers, 2010] for HMM estimation, `cvxpy` [Diamond and Boyd, 2016] for convex optimization, `scikit-learn` [Pedregosa et al., 2011] for Ledoit-Wolf shrinkage, and `joblib` for parallel fold execution. All code is available at `https://github.com/AbdelkrimCode/regime-adaptive-portfolio`. Continuous integration via GitHub Actions runs 55 unit and integration tests on every commit.

4 Results

4.1 Full-Period Performance (2006–2024)

Table 2 reports annualized performance metrics over the full backtest period. The regime-adaptive strategy achieves a Sharpe ratio of 0.453 versus 0.433 for SPY buy-and-hold — a modest improvement. The primary outperformance is in drawdown control: maximum drawdown of −26.1% versus −60.4% for SPY, and a Calmar ratio of 0.182 versus 0.137.

Table 2: Full-Period Performance (2006–2024)

Metric	Ours	SPY	EW	60/40	RP
Ann. Return	4.75%	8.27%	5.90%	6.80%	4.55%
Ann. Vol	7.53%	19.58%	11.19%	11.20%	7.58%
Sharpe	0.470	0.433	0.443	0.518	0.455
Max DD	-26.1%	-60.4%	-35.3%	-36.2%	-24.9%
Calmar	0.183	0.137	0.167	0.188	0.182

The strategy's volatility (7.53% annualized) is dramatically lower than all equity-heavy benchmarks, reflecting the regime-conditional allocation to bonds and gold during Bear and Crash periods. This lower volatility drives the Sharpe advantage despite lower raw returns.

4.2 Held-Out Test Period (2019–2024)

Table 3 reports performance on the held-out 2019–2024 period, which the walk-forward model never trained on as a whole. The strategy's Sharpe (0.291) lags SPY (0.677) significantly — the 2019–2024 period was characterized by a sustained equity bull market interrupted by brief, fast-recovering corrections, an environment unfavorable to regime-switching strategies.

Table 3: Held-Out Performance (2019–2024)

Metric	Ours	SPY
Ann. Return	4.55%	14.86%
Ann. Vol	8.45%	19.91%
Sharpe	0.291	0.677
Max DD	-26.1%	-35.75%
Calmar	0.175	0.416

4.3 Frozen Out-of-Sample Evaluation

To rule out lookahead in the walk-forward procedure itself, we train a single HMM on data through December 2018, freeze all parameters, and apply the model statically to 2019–2024 without any retraining. Table 4 compares frozen versus walk-forward performance.

Table 4: Frozen vs Walk-Forward (2019–2024)

Metric	Walk-Forward	Frozen	SPY
Ann. Return	4.55%	4.98%	14.86%
Ann. Vol	8.45%	8.71%	19.91%
Sharpe	0.291	0.324	0.677
Max DD	-26.1%	-25.5%	-35.75%
Calmar	0.175	0.195	0.416

The frozen model slightly outperforms walk-forward (0.324 vs 0.291 Sharpe). This confirms two things: (1) the strategy generalizes — performance does not depend on retraining within the test period; and (2) quarterly retraining at the test-period frequency adds noise rather than signal, consistent with the feature ablation finding that the baseline 3-feature model is already close to optimal.

4.4 Subperiod Analysis

Table 5 decomposes performance across three economically distinct subperiods.

Table 5: Subperiod Analysis

Period	Metric	Ours	SPY
GFC (2008–09)	Sharpe	-0.043	-0.345
	Max DD	-12.5%	-56.4%
	Calmar	-0.019	-0.274
Bull (2013–19)	Sharpe	0.739	1.003
	Max DD	-7.7%	-19.8%
	Calmar	0.562	0.669
COVID+Rates (2020–24)	Sharpe	0.070	0.528
	Max DD	-26.1%	-35.75%
	Calmar	0.104	0.331

The GFC result is the most compelling: maximum drawdown of −12.5% versus −56.4% for SPY, achieved without any forward-looking information. The strategy performs poorly in the COVID+rates period — the 2020 crash was too fast for quarterly retraining to adapt, and the 2022 environment (simultaneous equity and bond selloffs) violated the safe-haven assumption underlying the Crash optimizer.

4.5 Regime Distribution

Over the full period, the HMM assigns 1,309 days (28.3%) to Bull, 1,448 (31.3%) to Bear, 1,099 (23.8%) to Sideways, and 764 (16.5%) to Crash. Average regime duration ranges from 17.2 days (Bull) to 22.5 days (Crash), consistent with mean-reversion in market volatility. The empirical transition matrix shows Bear-to-Bull as the dominant transition (55.1%), reflecting the common pattern of sharp recoveries following drawdown periods.

5 Robustness Analysis

5.1 Bootstrap Significance Test

We test whether the strategy's Sharpe ratio exceeds SPY's using a block bootstrap [Politis and Romano, 1994] with block length 20 trading days to preserve autocorrelation structure. Under 1,000 bootstrap replications, the empirical p-value is 0.376 — the null hypothesis that the strategy's Sharpe is no better than SPY's cannot be rejected at conventional significance levels.

This result is reported prominently because it is honest: the strategy does not statistically outperform SPY on a risk-adjusted basis. The value proposition is drawdown control and tail risk reduction, not Sharpe maximization.

5.2 Walk-Forward Leakage Audit

We programmatically verify that no training fold observes future test data. For each of 74 quarterly folds, we record the last training date τ_{train} and the first test date τ_{test}. The condition $\tau_{\text{train}} < \tau_{\text{test}}$ holds for all 74 folds. Weekend and holiday offsets cause apparent one-to-two day gaps (e.g., training ends Friday, testing begins Monday), which are correct. Zero leakage is confirmed. The full audit is available at `results/walk_forward_audit.csv`.

5.3 Feature Ablation Study

Table 6 reports full pipeline results across four feature configurations. The baseline (21-day vol, 63-day correlation) is not cherry-picked: `vol_10` produces higher full-sample Sharpe but lower held-out Sharpe, while `skew_kurt` collapses out-of-sample despite reasonable in-sample performance. The baseline occupies a stable, robust region of the feature space.

Table 6: Feature Ablation Results

Feature Set	Full Sharpe	OOS Sharpe	Max DD
Baseline (vol_{21}, corr_{63})	0.453	0.291	-26.1%
vol_{10} (vol_{10}, corr_{63})	0.579	0.273	-23.9%
vol_{42} (vol_{42}, corr_{63})	0.410	0.310	-26.9%
skew+kurt	0.466	0.067	-28.1%

We additionally tested VIX and yield curve slope (5Y–3M Treasury spread) as supplementary macro features to reduce SPY circularity. Full-sample Sharpe improved to 0.558 but held-out Sharpe collapsed to 0.135 — the HMM learned VIX-regime associations specific to the 2006–2018 training period that did not generalize. The 3-feature baseline was restored.

5.4 Window Sensitivity Sweep

A 3×3 grid search over $\{10, 21, 42\}$ volatility windows and $\{42, 63, 126\}$ correlation windows yields full-sample Sharpe in the range $[0.41, 0.58]$ and held-out Sharpe in $[0.27, 0.31]$. The baseline configuration ($W_\sigma = 21$, $W_\rho = 63$) is not the in-sample optimum, further supporting that it is not cherry-picked.

5.5 Stress Tests

Table 7: Stress Test Results

Metric	Portfolio	SPY
VaR (95%, daily)	-0.77%	-1.83%
CVaR (95%, daily)	-1.17%	-3.06%
Sharpe, VIX > 30	-0.363	-2.000
MC Sharpe mean	0.463	—
MC Sharpe std	0.238	—
MC 95% CI	[-0.005, 0.924]	—

Daily VaR and CVaR at the 95th percentile are less than half those of SPY, consistent with the drawdown control result. Both strategies produce negative Sharpe during VIX > 30 periods — crisis environments are adverse for all strategies — but the portfolio's loss per unit of risk is substantially lower. Monte Carlo Sharpe (1,000 bootstrap paths) produces a mean of 0.463, closely matching the empirical estimate of 0.453, with no evidence of estimation bias.

5.6 PCA Regime Validation

Principal Component Analysis of the three features reveals that Bull, Bear, and Sideways regimes overlap heavily in the central region of PC1-PC2 space (39.2% and 33.3% variance explained respectively). Meaningful separation exists only in the left tail of PC1, corresponding to extreme negative returns and elevated volatility — Crash and severe Bear episodes.

This finding explains the non-significant bootstrap result: the HMM adds value primarily in tail events, not in distinguishing normal market conditions. Drawdown control is the correct framing for this strategy's contribution.

6 Limitations

Gaussian emission assumption. Jarque-Bera tests reject normality for all four regimes ($p \approx 0$). Bear regime kurtosis is 14.3 — extreme fat tails that Gaussian emissions cannot adequately model. A Student-t HMM [Zhang et al., 2004] would be more appropriate but is not supported by `hmmlearn`; implementation from scratch via EM with t-distribution updates is left for future work.

Markov assumption. HMMs assume the current regime depends only on the previous regime. Markets exhibit momentum, mean-reversion, and long-memory volatility clustering (GARCH effects) at multiple horizons, violating the first-order Markov property. A higher-order HMM or regime-duration model would capture these dynamics at the cost of significantly increased parameter count.

SPY circularity. All three features are derived from SPY returns. The HMM detects regimes using the same instrument it indirectly trades, introducing circularity in the signal. Macro features (VIX, yield curve slope) were tested as orthogonal alternatives but did not generalize out-of-sample. True signal orthogonality would require data from a different domain (e.g., economic survey indices, options market microstructure).

Expanding covariance window. The optimizer uses an expanding window of all available returns, growing from approximately 3,260 to 4,780 days over the backtest. Early folds are dominated by 2008 volatility ($\approx 60\%$ annualized); late folds mix this with 2024 volatility ($\approx 15\%$). A rolling window would be more stationary but introduces estimation instability in short windows.

Transaction cost model. The flat 2bp cost model does not account for market impact, bid-ask spread dynamics, or securities lending costs. For daily rebalancing at retail scale, realized costs would be higher. Sensitivity analysis over cost assumptions is available in the accompanying repository.

Time period bias. The 2006–2024 sample includes the GFC, a prolonged low-volatility bull market, COVID-19, and an aggressive rate cycle. The strategy has not been evaluated in structurally different environments such as the 1970s stagflation or prolonged deflation. Out-of-sample evidence from such periods is unavailable by construction.

Statistical significance. The bootstrap p-value of 0.376 confirms that excess Sharpe over SPY is not statistically significant. Any claim of outperformance should be framed around drawdown control — a claim supported by the data — rather than risk-adjusted returns.

7 Conclusion

We presented a regime-adaptive portfolio framework built on a causal Hidden Markov Model and regime-conditional convex optimization. The central methodological contribution is the replacement of standard HMM smoothing with a strictly forward-only filter, eliminating the within-window lookahead bias that inflates backtested performance in most published HMM portfolio systems.

Applied to eight liquid ETFs over 2006–2024, the strategy achieves a maximum drawdown of -26.1% versus -60.4% for SPY buy-and-hold — a 57% reduction in peak-to-trough loss. During the 2008–2009 Global Financial Crisis specifically, maximum drawdown is -12.5% versus -56.4% for SPY. A frozen out-of-sample evaluation confirms these results are not an artifact of walk-forward lookahead: a model trained exclusively on pre-2019 data achieves Sharpe 0.324 on the 2019–2024 period, comparable to the walk-forward estimate of 0.291.

The strategy does not statistically outperform SPY on a risk-adjusted basis (bootstrap $p = 0.376$), and underperforms significantly on raw returns in the 2019–2024 period — a sustained equity bull market unfavorable to regime-switching allocation. PCA analysis of the feature space confirms that regime separation is weak in normal market conditions and meaningful only in the left tail, explaining both the drawdown control result and the absence of Sharpe outperformance.

Future work includes replacing Gaussian emissions with Student-t distributions to better model fat-tailed returns, incorporating regime duration into the transition model, and extending the framework to higher-frequency data where the short-term predictive content of order book dynamics may provide more orthogonal signals for regime detection.

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References

Andrew Ang and Geert Bekaert. Regime changes in asset allocation. *Journal of Financial Economics*, 64(3):347–393, 2002.

Gérard Cornuéjols and Reha Tütüncü. *Optimization Methods in Finance*. Cambridge University Press, 2006.

Steven Diamond and Stephen Boyd. Cvxpy: A python-embedded modeling language for convex optimization. *Journal of Machine Learning Research*, 17(83):1–5, 2016.

Massimo Guidolin and Allan Timmermann. Asset allocation under multivariate regime switching. *Journal of Economic Dynamics and Control*, 31(11):3503–3544, 2007.

James D Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.

hmmlearn developers. hmmlearn: Hidden markov models in python. `https://hmmlearn.readthedocs.io`, 2010. Accessed 2024.

Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Dimitris N Politis and Joseph P Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

Christopher M Turner, Richard Startz, and Charles R Nelson. A markov model of heteroskedasticity, risk, and learning in the stock market. *Journal of Financial Economics*, 25 (1):3–22, 1989.

Ming Zhang, Gagan Bhatt, and Thomas Berger. A new approach to the student-t hidden markov model. *Pattern Recognition Letters*, 25(12):1423–1432, 2004.